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### System and method for shrimp cultivation

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#### Abstract

A system includes a housing; a water tank positioned within the housing, wherein an inside of the water tank is sealed to prevent contact between material forming the water tank and water within the water tank; a water movement subsystem causing water to circulate within the water tank, including a baffle extending along the center of the water tank dividing the water tank into a circular raceway, and a pump causing water to circulate; an aeration subsystem maintaining an oxygenation level suitable for crustaceans to live within the water tank; a temperature control subsystem maintaining a water temperature suitable for crustaceans to live within the water tank; a water quality monitoring subsystem monitoring a water quality parameter of water within the water tank; a feeding subsystem configured to dispense food into the water tank; a biofloc removal subsystem configured to remove biofloc from the water tank; and a control system.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation application of U.S. application Ser. No. 18/178,617, filed Mar. 6, 2023, which is a continuation application of International (PCT) Patent Application No. PCT/US2021/054263, filed Oct. 8, 2021, which relates to and claims the benefit of commonly-owned, U.S. Provisional Patent Application No. 63/089,206, filed on Oct. 8, 2020 and entitled "SYSTEM AND METHOD FOR SHRIMP CULTIVATION," the contents of which are incorporated herein by reference in their entirety.

### FIELD OF THE INVENTION

(1) The field of invention relates to systems and method for cultivation of crustaceans. More particularly, the field of invention relates to systems and methods for cultivation of shrimp requiring minimal user interaction.

### BACKGROUND OF THE INVENTION

(2) Shrimp and other crustaceans are commonly used as food. Due to sustainability concerns relating to harvesting of wild populations, farmed production of such animals is desirable. However, existing farming techniques suffer from low efficiency, involve use of harmful substances and excessive use of natural resources, and require significant user interaction.

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## Description

### BRIEF DESCRIPTION OF THE FIGURES

(1) Some embodiments of the invention are herein described, by way of example only, with reference to the accompanying drawings. With specific reference now to the drawings in detail, it is stressed that the particulars shown are by way of example and for purposes of illustrative discussion of embodiments of the invention. In this regard, the description taken with the drawings makes apparent to those skilled in the art how embodiments of the invention may be practiced.

(2) FIG. 1 shows an exemplary system for farm-raising shrimp.

(3) FIG. 2A shows a first exemplary schematic illustration of the elements of an exemplary system.

(4) FIG. 2B shows a second exemplary schematic illustration of the elements of an exemplary system.

(5) FIG. 2C shows an exemplary software control block diagram of an exemplary system.

(6) FIG. 2D shows an exemplary software control block diagram of an exemplary system.

(7) FIG. 2E shows elements of an exemplary system.

(8) FIG. 3A shows the general layout of space within an exemplary system.

(9) FIG. 3B shows the general layout of structural elements of an exemplary system.

(10) FIG. 4A shows a first exemplary main aeration subsystem.

(11) FIG. 4B shows a second exemplary main aeration subsystem.

- (12) FIG. 5A shows an exemplary water movement subsystem.
- (13) FIG. 5B shows an exemplary drainage subsystem.
- (14) FIG. 5C shows an exemplary solids collection box.
- (15) FIG. 6A shows a first exemplary waste collection subsystem.
- (16) FIG. 6B shows a second exemplary waste collection subsystem.
- (17) FIG. 7A shows an exemplary sensor unit.
- (18) FIG. 7B shows an exemplary water quality monitoring subsystem.
- (19) FIG. 8A shows an exemplary feeding subsystem.
- (20) FIG. 8B shows an exemplary feeding subsystem.
- (21) FIG. 9 shows an exemplary temperature control subsystem.
- (22) FIG. 10A shows an exemplary biofloc removal subsystem.
- (23) FIG. 10B shows an exemplary decanter of an exemplary biofloc removal subsystem.

#### SUMMARY OF THE INVENTION

(24) The exemplary embodiments relate to systems for farm-raising of shrimp, and related farming methods. In an embodiment, a system is self-contained. In an embodiment, hardware elements of a system are self-contained and are controlled by software and/or user controls that are located remotely, such as by “cloud” software. In an embodiment, hardware elements of a system are self-contained within a housing. In an embodiment, the housing is a shipping container of the type referred to as an “intermodal container”.

(25) In some embodiments, a system includes a housing containing one or more water tanks; a water movement system operable to cause water to circulate within the one or more water tanks; a main aeration system operable to maintain a desired oxygenation level within the one or more water tanks; a drain subsystem operable to remove waste material from the one or more water tanks; a temperature control subsystem operable to maintain a desired water temperature within the one or more water tanks; a water quality monitoring subsystem configured to monitor one or more water quality parameters of water within the one or more water tanks; and a feeding subsystem operable to dispense food into the one or more water tanks.

(26) In some embodiments, the housing includes a shipping container. In some embodiments, the one or more water tanks includes two water tanks.

(27) In some embodiments, a system includes a housing, wherein the housing is an intermodal container; at least one water tank positioned within the housing, wherein an inside of each of the at least one water tank is sealed so as to prevent contact between a material forming the at least one water tank and water within the at least one water tank; a water movement subsystem operable to cause water to circulate within the at least one water tank, wherein the water movement system comprises: at least one baffle extending longitudinally along a portion of a center of the at least one water tank so as to divide the at least one water tank into a generally circular raceway; and at least one pump configured to cause water to circulate about the at least one baffle; an aeration subsystem configured to maintain an oxygenation level within the at least one water tank, wherein the oxygenation level is suitable for crustaceans to live within water in the at least one water tank; a temperature control subsystem operable to maintain a water temperature within the at least one water tank, wherein the water temperature is suitable for crustaceans to live within water in the at least one water tank; a water quality monitoring subsystem configured to monitor at least one water quality parameter of water within the at least one water tank, wherein the at least one water quality parameter includes a temperature, a dissolved oxygen concentration, a nitrogen concentration, a phosphate concentration, a pH, and a salinity; a feeding subsystem configured to dispense food into the at least one water tank; a biofloc removal subsystem configured to remove biofloc from the at least one water tank; and a computer-operated control system configured to operate the water movement system, the aeration system, the temperature control subsystem, the water quality monitoring subsystem, and the feeding subsystem.

(28) In some embodiments, the aeration subsystem includes at least one main air tube extending

above each of the at least one water tank, and a plurality of branch tubes extending away from each of the at least one main air tube into water within the at least one water tank, wherein each of the plurality of branch tubes includes aeration tubing. In some embodiments, each of the plurality of branch tubes further includes PVC tubing. In some embodiments, the PVC tubing has a diameter of  $\frac{1}{2}$  inch. In some embodiments, the aeration tubing includes porous tubing having an average pore size that is in a range of from 0.001 inch to 0.004 inch. In some embodiments, the aeration subsystem includes a main aeration subsystem and a secondary aeration subsystem. In some embodiments, the secondary aeration subsystem includes at least one venturi eductor coupled to the water movement subsystem.

(29) In some embodiments, the feeding subsystem includes a scale; a feed container positioned on the scale; a valve positioned at a bottom of the feed container and operable to allow feed to pass therethrough when the valve is in an open position; and a feeding tube coupled to the aeration subsystem and the valve such that air provided by the aeration subsystem propels feed passing through the valve into the at least one water tank. In some embodiments, the system includes one of the feeding subsystem for each of the at least one water tank.

(30) In some embodiments, the at least one water tank includes a first water tank positioned at a bottom of the housing and a second water tank positioned above the first water tank. In some embodiments, the system also includes a support structure configured to support weight of at least the second water tank.

(31) In some embodiments, each of the at least one water tank has a volume that is in a range of from 10 cubic meters to 50 cubic meters.

(32) In some embodiments, the system also includes a control room positioned at a first end of the housing.

(33) In some embodiments, the system is configured to support presence of biofloc within the at least one water tank. In some embodiments, the computer-operated control system is configured to support presence of biofloc within the at least one water tank by controlling a ratio of carbon to nitrogen within water within the at least one water tank. In some embodiments, the computer-operated control system is configured to support presence of biofloc within the at least one water tank by controlling the ratio of carbon to nitrogen within water within the at least one water tank to be in a range of from 12:1 to 20:1.

(34) In some embodiments, the biofloc removal subsystem includes a decanter positioned within each of the at least one water tank such that excess water within the decanter overflows into the at least one water tank, wherein the decanter is shaped such that biofloc within water in the decanter settles to a bottom of the decanter; a pump configured to pump water from each of the at least one water tank and into the decanter; a drain pump operable to remove biofloc from the bottom of the decanter when the drain pump is activated; and a biofloc aeration arrangement operable to cause air to flow into the bottom of the decanter when the biofloc aeration arrangement is activated, thereby to cause biofloc to mix within water within the decanter and to overflow into the at least one water tank. In some embodiments, the biofloc aeration arrangement includes a valve that is operable to activate the aeration arrangement; and an aeration tube that is coupled to the valve and to the aeration subsystem.

(35) In some embodiments, the system also includes a drain subsystem operable to remove waste material from the at least one water tank.

(36) In some embodiments, each of the at least one water tank is sealed by a water-resistant paint.

#### DETAILED DESCRIPTION OF THE INVENTION

(37) The exemplary embodiments relate to systems for farm-raising of shrimp, and related farming methods. In an embodiment, a system is self-contained. In an embodiment, hardware elements of a system are self-contained and are controlled by software and/or user controls that are located remotely, such as by “cloud” software. In an embodiment, hardware elements of a system are self-contained within a housing. In an embodiment, the housing is a shipping container of the type

referred to as an “intermodal container”. FIG. 1 shows a rendering of an exemplary system **100**.

(38) In some embodiments, the exemplary system **100** is configured to raise and harvest crustaceans such as shrimp through biofloc culture. It will be known to those of skill in the art that biofloc aquaculture describes an aquaculture practice in which waste materials (e.g., unused food and excreta) are converted to a protein-rich live feed by microbes present within the water. It will be further known that the term “biofloc” refers to a heterogeneous aggregate of organisms such as microalgae, bacteria, protozoa, zooplankton, and nematodes, as well as feces and uneaten food, which is typically held together by mucus secreted by bacteria, and which typically range in size from 50 to 200 microns. In some embodiments, use of biofloc aquaculture enables shrimp to be produced in a sustainable manner.

(39) In some embodiments, an exemplary system **100** includes a retrofitted 12-meter-long shipping container **110** (e.g., a housing) that has been subdivided to include two tanks, each 10 meters long and 80 centimeters deep, and positioned one over the other, with a control room **120** provided in the remaining space at a first end of the shipping container **110**. In some embodiments, an exemplary system **100** includes hatches to provide access to each tank.

(40) FIG. 2A shows a first exemplary schematic block diagram **200** of the exemplary system **100**. FIG. 2B shows a second exemplary block diagram **250** of the exemplary system **100**. In some embodiments, the system **100** includes a main controller **210** (e.g., a computer-operated control system including a combination of hardware and software) that has internet access (e.g., via a WiFi connection) enabling the controller **210** to be controlled by algorithms operating either locally or on a remote server. In some embodiments, the control algorithms are operative to receive monitoring data (e.g., as described herein) and provide an indication of water quality. In some embodiments, the control algorithms are operable to trigger a corrective action based on the indication of water quality. In some embodiments, the controller can operate autonomously in accordance with established instructions in the event of lost connectivity with the server. In some embodiments, flows of water and air are controlled by automatic ball valves **220** that are operable by the main controller **210** in order to allow the control algorithms to control the various settings that are needed during the production cycle. For clarity, only one of the ball valves **220** is specifically identified in each of FIG. 2A and FIG. 2B. In some embodiments, the automatic ball valves **220** are of the type commercialized by U.S. Solid of Cleveland, Ohio. In some embodiments, the system **100** includes a main panel having a contactor section that interfaces the main controller with devices that need electrical power. In some embodiments, all valves **220**, motors, etc., are located in the control room **120**. FIG. 2C shows an exemplary software control block diagram **260** including software running on the controller **210** and connections from the controller **210** to the remaining elements of the system **100**, as described herein. FIG. 2D shows an exemplary software control block diagram **270** including software running on the controller **210** and connections from the controller **210** to the remaining elements of the system **100**, as described herein. FIG. 2E shows a rendering of the exemplary system **100** with the shipping container **110** not shown to allow internal elements of the system **100** to be seen.

(41) FIG. 3A shows a general schematic view of the system **100** with the shipping container **110** not shown, allowing view of the major structural elements of the system **100**. As shown in FIG. 3A, the system **100** includes a control room **120** as discussed above and water tanks **310**. In some embodiments, the system **100** includes an upper water tank **312** and a lower water tank **314**. Throughout this disclosure, the term “water tank **310**” will be used to refer generically to either water tank and related elements, while the terms “upper water tank **312**” and “lower water tank **314**” will be used to refer to a specific one of the water tanks. FIG. 3B shows a general schematic view of structural elements of the system **100** with the shipping container **110** not shown. As shown in FIG. 3B, the system **100** includes a support structure **320** supporting the water tanks **310**. In some embodiments, the support structure **320** includes a reinforcement of the main structure of the shipping container **110**. In some embodiments, the support structure **320** includes a base to support

the upper water tank **312**.

(42) In some embodiments, production of shrimp begins with shrimp at the postlarva stage of the life cycle (e.g., at an average weight of less than 10 milligrams). In some embodiments, shrimp postlarva are those commercialized by Maricultura Vigas Sapi de CV of Lerma, Campeche, Mexico. In some embodiments, shrimp postlarva are free of disease. In some embodiments, juvenile shrimp are introduced into the exemplary system **100**. In some embodiments, the juvenile shrimp used in the exemplary system **100** have an average weight of 1 gram. In some embodiments, juvenile shrimp are provided into each water tank **310** (e.g., the lower tank **314** and the upper tank **312**) of the exemplary system at an average density of 400 shrimp per square meter of plan view area. In some embodiments, each water tank **310** is 10 meters in length and 2.4 meters in width, yielding a plan view area of 24 square meters. In some embodiments, each water tank **310** has a depth of 0.8 meters, yielding a useful volume of 19.2 cubic meters. In some embodiments, each tank has a useful volume that is in a range of 10 cubic meters to 50 cubic meters. In some embodiments, each tank has a useful volume that is in a range of 10 cubic meters to 40 cubic meters. In some embodiments, each tank has a useful volume that is in a range of 10 cubic meters to 30 cubic meters. In some embodiments, each tank has a useful volume that is in a range of 5 cubic meters to 50 cubic meters. In some embodiments, each tank has a useful volume that is in a range of 5 cubic meters to 40 cubic meters. In some embodiments, each tank has a useful volume that is in a range of 5 cubic meters to 30 cubic meters. In some embodiments, each tank has a useful volume that is in a range of 12 cubic meters to 28 cubic meters. In some embodiments, each tank has a useful volume that is in a range of 14 cubic meters to 26 cubic meters. In some embodiments, each tank has a useful volume that is in a range of 15 cubic meters to 25 cubic meters. In some embodiments, each tank has a useful volume that is in a range of 16 cubic meters to 24 cubic meters. In some embodiments, each tank has a useful volume that is in a range of 17 cubic meters to 23 cubic meters. In some embodiments, each tank has a useful volume that is in a range of 18 cubic meters to 22 cubic meters. In some embodiments, each tank has a useful volume that is in a range of 19 cubic meters to 21 cubic meters. In some embodiments, each tank has a useful volume that is in a range of 19 cubic meters to 20 cubic meters. In some embodiments, each tank has a useful volume that is in a range of 19 cubic meters to 19.5 cubic meters.

(43) In some embodiments, each water tank **310** is linked with a geomembrane. In some embodiments, each water tank **310** is lined with high density polyethylene (“HDPE”) having a thickness of 0.8 mm. In some embodiments, each water tank **310** includes a diffuse air aeration system, in which air is injected into the system through a blower and aeration tubes, as will be described in further detail hereinafter. In some embodiments, each water tank **310** includes a baffle (e.g., an artificial substrate) that increases the surface area of each water tank **310**. In some embodiments, each baffle is 8 meters wide and 0.8 meters high (e.g., is sufficiently tall to span the entire depth of the water in the tank). In some embodiments, each baffle has a surface area of 6.4 square meters per side, and 12.8 square meters total for both opposing sides. In some embodiments, each baffle is made from a felt material such as the type of material used to line automobile floors (e.g., the material commercialized under part number #MTCARPE20000030GRE01UA by Goldwheel USA Inc. of Irwindale, California. In some embodiments, shrimps will be fed for 24 hours/day, using automatic pneumatic feeders.

(44) In some embodiments, the ratio of carbon to nitrogen within the water of the water tanks **310** is controlled to accelerate the formation of bioflocs in the water. In some embodiments, the ratio of carbon to nitrogen is controlled to be 16:1, or to be about 16:1, or to be between 15:1 and 17:1, or to be between 14:1 and 18:1, or to be between 13:1 and 19:1, or to be between 12:1 and 20:1. In some embodiments, the ratio of carbon to nitrogen is controlled through organic fertilization. In some embodiments, organic fertilization uses sugar as a carbon source.

(45) In some embodiments, a water quality management process includes monitoring of water quality parameters including temperature, dissolved oxygen concentration, nitrogen concentration,

phosphate concentration, pH, and salinity of water within each water tank **310** (e.g., the upper tank **312** and the lower tank **314**). In some embodiments, water quality parameters are monitored by automatic probes that are connected to both tanks. In some embodiments, the automatic probes provide real-time measurements of water quality parameters. In some embodiments, water quality parameters are measured daily. In some embodiments, pH is measured by a titration method such as American Public Health Association Method 2320. In some embodiments, an optimal pH for the growth of shrimp is between 7.5 and 8.5. However, consumption of inorganic carbon by autotrophic bacteria present in the biofloc can cause a decrease in the pH of the water. To remedy this decrease in pH, in some embodiments, if correction of pH is needed, hydrated lime (e.g., calcium hydroxide) is added to the water. In some embodiments, the hydrated lime is added at a concentration of 0.05 grams per liter. In some embodiments, correction of the pH of the water allows the water in the tanks to be used for many cycles, thereby contributing to the zero-waste nature of the exemplary system.

(46) In some embodiments, nitrogen levels are evaluated by monitoring daily ammonia levels. In some embodiments, nitrogen levels are evaluated by monitoring daily nitrite levels. In some embodiments, phosphate levels are evaluated by monitoring orthophosphate levels on a weekly basis.

(47) In some embodiments, the concentration of total suspended solids is periodically determined. In some embodiments, the concentration of total suspended solids is determined on a weekly basis. In some embodiments, the concentration of total suspended solids is kept at 500 milligrams per liter. In some embodiments, excess suspended solids are the concentration of total suspended solids is controlled through the use of filters as will be described in greater detail hereinafter.

(48) In some embodiments, the volume of sedimentable flakes is periodically quantified. In some embodiments, the volume of sedimentable flakes is quantified three times per week. In some embodiments, the volume of sedimentable flakes is quantified through use of an Imhoff cone. In some embodiments, a typical acceptable range of volume of sedimentable flakes is from 10 to 15 milliliters per liter. In some embodiments, if the volume of sedimentable flakes exceeds an acceptable range, the system is operated so as to clarify the solids (e.g., remove some of the sedimentable flakes by use of the waste collection subsystem).

(49) In some embodiments, a probiotic is periodically applied to help maintain the quality of the water and the health of the shrimp. In some embodiments, the probiotic used is the probiotic commercialized under the trade name SANOLIFE by INVE Aquaculture of Salt Lake City, Utah. In some embodiments, the probiotic is applied directly to the water. In some embodiments, the probiotic is applied directly to the water two times per week. In some embodiments, the probiotic is applied directly to the water at a concentration of 0.5 ppm. In some embodiments, the probiotic is applied by mixing in with food. In some embodiments, the probiotic is applied by mixing in with food on a daily basis. In some embodiments, the probiotic is applied by mixing in with food at a concentration of 3 grams of probiotic per kilogram of food. In some embodiments, the probiotic is applied both by application directly to the water and by mixing in with food.

(50) In some embodiments, at the outset of cultivation, an initial biometry is performed to estimate the average weight of shrimp to be cultivated. In some embodiments, the biometry is performed on a sample size of 100 shrimp. In some embodiments, during the cultivation period, biometry is periodically performed. In some embodiments, the periodic biometry is performed weekly. In some embodiments, the periodic biometry is performed on a randomly selected sample from each tank. In some embodiments, the sample size is 100 shrimp. In some embodiments, the sample is weighed. In some embodiments, each individual shrimp in the sample is weighed. In some embodiments, weighing is performed using a digital scale having an accuracy of 0.01 grams. In some embodiments, an average weight of the shrimp is calculated based on the measured weights of all sampled shrimp. In some embodiments, the amount of feed to be provided is adjusted on a weekly basis based on the average weight.



(51) In some embodiments, to provide a safe environment for the growing of the organisms, exemplary systems include regenerative blowers and venturi tubes to maintain a good oxygen level in the water, centrifugal pumps to move water through the different components of the exemplary system, automatic feeders to provide a consistent and regular feeding schedule, monitoring equipment to inform control software (which will be described hereinafter) of levels of water quality variables, and a waste recollection system to collect feeding leftovers, skin shedding, etc. In some embodiments, a desirable oxygen level is at least 4 milligrams per liter. In some embodiments, a desirable oxygen level is at least 5 milligrams per liter. In some embodiments, these elements are positioned within the control room. In some embodiments, the control room includes an electrical load center capable of supplying sufficient electrical power to power these elements. In some embodiments, the control room includes an automatic power backup system in case of lack of utility power.

(52) In some embodiments, the retrofitting of a standard shipping container to produce an exemplary system includes reinforcing the main structure of the shipping container to support the weight of both tanks, construction of a base (e.g., a metal base, although any other material capable of supporting sufficient weight could also be used) to support the upper tank **312**, and construction of a dividing wall to separate the control room from the tanks. In some embodiments, existing container doors are used to provide access to the control room. In some embodiments, hatched openings are constructed to provide access to the upper tank **312** and the lower tank **314** to personnel. In some embodiments, the system includes one or more hatches on the top of the shipping container to provide access to the upper tank **312**. In some embodiments, the system includes one or more hatches on the side(s) of the shipping container to provide access to the lower tank **314**. In some embodiments, each water tank **310** is 10 meters long, 2.4 meters wide, and 0.8 meters deep. In some embodiments, each water tank **310** is made from a metal (or other material capable of providing sufficient structural strength). In some embodiments, each water tank **310** is sealed to prevent direct contact between the metal forming the water tank **310** and water within the water tank **310**. In some embodiments, each water tank is sealed by lining with a geo-liner (e.g., a geomembrane including a material such as HDPE, linear low-density polyethylene, polyvinyl chloride, flexible polypropylene, chlorosulfonated polyethylene, or ethylene propylene diene terpolymer). In some embodiments, each water tank **310** is coated by an epoxy. In some embodiments, each water tank **310** is not lined. In some embodiments, the inside of each water tank **310** is painted so as to seal the water tank **310**. In some embodiments, the inside of each water tank **310** is painted with a paint that is suitable to separate the metal forming the water tank **310** from the water within the water tank **310**. In some embodiments, the inside of each water tank **310** is painted with a water-resistant paint that is suitable to separate the metal forming the water tank **310** from the water within the water tank **310**. In some embodiments, the paint does not include hexavalent chromium or other mutagenic components. In some embodiments, each water tank **310** includes a waste collection subsystem, which includes a pumping system and a filtering system.

(53) In some embodiments, an exemplary system includes at least one water pump providing water movement within the exemplary system. In some embodiments, the at least one water pump is capable of providing sufficient pressure to fill and drain the water tanks **310** as needed. In some embodiments, the at least one water pump is capable of providing sufficient water pressure for a supplementary oxygenation venturi system (described in further detail hereinafter). In some embodiments, the at least one water pump is capable of providing sufficient water pressure to provide circular movement to the water mass in order to allow for the collection of solids within the water mass. In some embodiments, the at least one water pump includes a centrifugal water pump. In some embodiments, the at least one water pump includes a centrifugal water pump having a power that is in a range of between 0.5 horsepower and 3 horsepower. In some embodiments, the at least one water pump includes a 2-horsepower centrifugal water pump. In some embodiments, the at least one water pump includes a 1.5 horsepower centrifugal water pump. In some

embodiments, the at least one water pump is coupled to a piping arrangement that connects the at least one water pump to the different elements of the system (e.g., the tanks, the venturi system, etc.) referenced above). In some embodiments, the at least one water pump and any valves in the piping arrangement are positioned within the control room. In some embodiments, the at least one water pump is coupled to a jet array configured to impart a circular movement to the water in the water tanks **310**, thereby enabling waste collection.

(54) In some embodiments, the exemplary system **100** includes a main aeration subsystem capable of providing a level of aeration within the tanks that is sufficient for cultivation of shrimp. In some embodiments, a sufficient oxygen level is at least 4 milligrams per liter. In some embodiments, a sufficient oxygen level is at least 5 milligrams per liter. FIG. **4A** shows a first exemplary main aeration subsystem **400**. In some embodiments, the main aeration subsystem **400** includes a regenerative blower **410**. In some embodiments, the regenerative blower **410** is a 1.5 horsepower regenerative blower. In some embodiments, the regenerative blower **410** is a 1 horsepower regenerative blower. In some embodiments, the regenerative blower **410** is coupled to a network of tubes. In some embodiments, the network of tubes includes a main tube **420** running parallel to each tank **310**. In some embodiments, the network of tubes includes a plurality of smaller branch tubes **430** extending away from each main tube **420** and across each tank **310**. In some embodiments, each of the smaller branch tubes **430** includes a first portion **432** extending away from the corresponding main tube **420** and downward into the tank **310**, and a second portion **434** extending away from the first portion **432** and across the tank **310**. In some embodiments, the second portions **434** include aeration tubing.

(55) FIG. **4B** shows a second exemplary main aeration subsystem **450**. In some embodiments, the main aeration subsystem includes a regenerative blower **460**. In some embodiments, the regenerative blower **460** is a 1.5 horsepower regenerative blower. In some embodiments, the regenerative blower **460** is a 1 horsepower regenerative blower. In some embodiments, the regenerative blower **460** is coupled to a network of tubes. In some embodiments, the network of tubes includes two main tubes **470** for each tank **310**. In some embodiments, the two main tubes **470** run above the tanks **310** along the length of each tank **310**, with the two main tubes **470** for each tank running along opposite sides of each tank **310**. In some embodiments, a plurality of smaller branch tubes **480** extends away from each of the main tubes. In some embodiments, each of the branch tubes **480** includes a first portion **482** that extends downward into each tank and a second portion **484** that extends along a portion of the length of the tank **310**. In some embodiments, the second portions **484** of the branch tubes **480** are positioned along the floor of the tanks **310**. In some embodiments, the second portions **484** include aeration tubing.

(56) In some embodiments, some or all of the main tubes and branch tubes are made from aeration tubing. In some embodiments, some or all of the main tubes and branch tubes are made from porous plastic aeration tubing. In some embodiments, some or all of the main tubes and branch tubes are made from porous plastic aeration tubing including rubber and polyethylene. In some embodiments, some or all of the main tubes and branch tubes are made from porous plastic aeration tubing having pores of an average diameter that is in the range of 0.001 inches to 0.004 inches. In some embodiments, some or all of the main tubes and branch tubes are made from the aeration tubing commercialized under the trade name AERO-TUBE by Swan Products LLC of Marion, Ohio. In some embodiments, the aeration tubing is operable to provide microbubbles of oxygen to aerate the water. In some embodiments, some or all of the tubes include sections of PVC tubing interspersed with sections of aeration tubing. In some embodiments, the branch tubes include ½-inch diameter PVC tubing and aeration tubing.

(57) In some embodiments, the system **100** includes a secondary aeration subsystem that is capable of supplementing the main aeration subsystem in the event that the oxygen level within the tanks is insufficient. In some embodiments, secondary aeration includes routing water from the circulation pump through the venturi tubes to provide additional aeration. It will be understood by those of

skill in the art that venturi tubes use the differential pressure potential in a pipe to create a vacuum and suction from a second pipe. In the present case, in some embodiments, pressurized water is passed through a venturi manifold, thereby creating suction in a second tube that reaches the surface of the water and causing air to be pulled through the second tube. In some embodiments, the air is then mixed with the water and expelled at the output of the venturi as water mixed with fine air bubbles.

(58) In some embodiments, the exemplary system includes a water movement subsystem capable of causing the water contained within the tanks to circulate therein. FIG. 5A shows an exemplary water movement subsystem **500**. In some embodiments, the water movement subsystem **500** includes a baffle **510** positioned within each tank **310** and extending longitudinally along the center of each tank **310**. In some embodiments, the baffle **510** changes the effective shape of the tank **310**, causing the tank **310** to effectively be shaped as a circular raceway. In some embodiments, the baffle **510** is also configured to allow shrimp to attach thereto. In some embodiments, the baffle **510** is made from a synthetic material. In some embodiments, the baffle **510** is made from a material having a rough surface. In some embodiments, the rough surface of the baffle **510** allows shrimp to attach to the baffle **510** and allows acceleration of nitrifying bacteria. In some embodiments, the baffle **510** is made from car mat fabric. In some embodiments, the water movement subsystem includes an array of jets **520** positioned along the baffle **510**. In some embodiments, the jets **520** point in opposite directions (e.g., jets **520** positioned on a first side of the baffle **510** are oriented to force water to flow away from the control room, while jets **520** positioned on an opposing second side of the baffle **510** are oriented to force water to flow toward the control room), thereby to impart a circular movement to the water when the jets **520** are active. In some embodiments, the water movement subsystem **500** can be turned on and off as needed by the main controller. For example, in some embodiments, the water movement subsystem **500** is turned on when it is necessary to collect solids (e.g., waste) from the bottom of the tank. In some embodiments, the water movement subsystem **500** is engaged to cause circulation at timed intervals. In some embodiments, the timed intervals are determined based on the feeding schedule. In some embodiments, the timed intervals are set to be a certain amount of time after feeding occurs in order to allow the shrimp sufficient time to feed and to determine the amount of leftover feed. In some embodiments, the water movement subsystem **500** is controlled by an automated valve that is configured to choose between normal or water movement recirculation. In some embodiments, based on requirements at a given time, the water movement subsystem **500** is configured to provide either: (1) normal recirculation, which does not impart circular movement to the water, but which provides sufficient movement to circulate water through the heat exchangers (described below) and allow for temperature control; (2) water movement recirculation, which imparts circular movement to the water; or (3) extra oxygenation, which cycles water through the venturi tubes for additional oxygenation, as discussed above.

(59) FIG. 5B shows an exemplary drainage subsystem **550**. In some embodiments, the drainage subsystem **550** operates in cooperation with the water movement supplied by the water movement subsystem **500**. In some embodiments, the drainage subsystem **550** includes a solids collection box **560** in the bottom of each water tank **310**. FIG. 5C shows various views of an exemplary solids collection box **560**. In some embodiments, water movement supplied by the water movement subsystem **500** causes solids to circulate within each water tank **310** and settle into the solids collection box **560**. In some embodiments, the drainage subsystem **550** includes a valve **570** (e.g., a ball valve) at the bottom of each solids collection box **560**. In some embodiments, the valve **570** is operable (e.g., selectively by an operator) to drain the corresponding water tank **310** (including settled solids and/or water) through a pipe **580**. In some embodiments, each pipe **580** drains to the exterior of the system **100** (e.g., outside the shipping container). In some embodiments, shrimp can be harvested by using a harvesting net on the output of each pipe **580**.

(60) In some embodiments, an exemplary system **100** includes a waste collection subsystem that is

configured to collect solid material (e.g., waste) that sinks to the bottom of the tanks **310** when water is stationary, and which is caused to circulate by the exemplary water movement subsystem **500** or **550** described above. FIG. **6A** shows a first exemplary waste collection subsystem **600**. In some embodiments, the exemplary waste collection subsystem **600** includes waste collectors **610** positioned within each tank. In some embodiments, the waste collectors **610** include a filter configured to remove waste from water passing therethrough. In some embodiments, the filters are interchangeable. In some embodiments, the waste collection subsystem **600** also includes a solids pump **620** configured to remove waste from the waste collectors **610**. In some embodiments, the waste collection subsystem **600** includes a single solids pump **620** that is configured to remove waste from all of the waste collectors **610**; in other embodiments, the waste collection subsystem **600** includes a separate solids pump **620** for each of the waste collectors **610**. In some embodiments, waste that is being removed from the tanks by the waste collection subsystem **600** can be sampled. In some embodiments, the contents of sampled waste can provide insights into the health of the shrimp, effectiveness of the feeding strategy, shredding, etc. In some embodiments, the waste collection subsystem **600** can also be used to extract biofloc from the tanks if necessary. In other embodiments, as will be described hereinafter, the system **100** includes a biofloc removal subsystem that is separate from the waste collection subsystem **600**. In some embodiments, the waste collection subsystem **600** can also be used to empty and drain the tanks **310**. In some embodiments, a harvesting system can be coupled to the waste collection subsystem to harvest shrimp from the tanks.

(61) FIG. **6B** shows a second exemplary waste collection subsystem **650**. In some embodiments, the waste collection subsystem **650** includes waste collectors **655** formed in the bottom of each tank **310**. In some embodiments, The circular motion created by the water movement subsystem **500** allows material that sinks to the bottom of each tank **310** to be collected within the waste collectors **655**. In some embodiments, the waste collectors **655** are periodically emptied of solid material by solid waste collection pumps **660** causing waste to be emptied by solid waste drainage tubes **665**. In some embodiments, the waste collection subsystem **650** includes water waste pumps **670** that pump water to filters **675** with exchangeable screens that can be opened to sample the material at the bottom of the tanks **310** which can give insights into the health of the organisms, effectiveness of the feeding strategy, shredding, etc. In some embodiments, each tank **310** has its own solid waste collection pump **660** and its own water waste pump **670**. In some embodiments, water is removed from each waste collector **655**, pumped through the filters **675** by the pumps **670**, and returned to the tanks via water tubes **680**. In some embodiments, waste is removed via waste tubes. In some embodiments, the pumps **660** are activated at programmable intervals throughout the day. In some embodiments, the filter is emptied once a day by an operator to sample the amount and quality of the waste matter (e.g. number of dead, weight, etc.).

(62) In some embodiments, the exemplary system includes a water quality monitoring subsystem. In some embodiments, the water quality monitoring subsystem **700** includes a sensor unit positioned within each tank. FIG. **7A** illustrates an exemplary sensor unit **710**. In some embodiments, the exemplary sensor unit **710** includes sensors operable to detect water parameters such as temperature, dissolved oxygen concentration, nitrogen concentration, phosphate concentration, pH, salinity, etc. In some embodiments, the exemplary sensor unit **710** includes a communication interface capable of transmitting, and configured to transmit, data detected thereby to the controller **210**, thereby to enable the controller **210** to initiate appropriate interventions.

(63) FIG. **7B** shows another exemplary water quality monitoring subsystem. In some embodiments, the water quality monitoring subsystem **700** includes a sampling element **720** positioned in each water tank **310**. In some embodiments, each sampling element **720** is configured to sample water within the corresponding water tank **310** and provide the sampled water to an analysis element **730** positioned outside the water tanks **310**. In some embodiments, the analysis element **730** includes sensors operable to detect water parameters such as temperature, dissolved oxygen concentration,

nitrogen concentration, phosphate concentration, pH, salinity, etc. In some embodiments, the analysis element **730** includes a communication interface capable of transmitting, and configured to transmit, data detected thereby to the controller **210**, thereby to enable the controller **210** to initiate appropriate interventions.

(64) In some embodiments, the exemplary system includes a feeding subsystem. FIG. **8A** shows an exemplary feeding subsystem **800**. In some embodiments, the exemplary feeding subsystem **800** includes a container **810** for feed that is positioned atop a weight sensor **820**, thereby enabling precise amounts of food to be dispensed. In some embodiments, the container **810** is opened and closed by a motorized ball valve **830**, as described above, thereby enabling the container **810** to be opened and closed by the main controller of the exemplary system. In some embodiments, the container **810** is positioned above the outlet of the regenerative blower **410** or **460** of the main aeration subsystem **400** or **450**. In some embodiments, the container **810** is positioned such that the pressurized air output by the regenerative blower forces the feed into a venturi eductor **840** and subsequently into the tank. In some embodiments, as a result of use of a venturi eductor **840**, feed is encouraged to fall into the feeding tube. In some embodiments, the feeding subsystem **800** is configured such that the main controller **210** can open the intake from the regenerative blower **410** or **460** at desired intervals to thereby propel feed into the tanks. In some embodiments, the feeding subsystem **800** is configured to selectively feed either the top tank **310** or the bottom tank **310**. In some embodiments, the exemplary system includes two exemplary feeding subsystems **800**, a first one of which is configured to feed the top tank **310** and a second one of which is configured to feed the bottom tank **310**. In some embodiments, the regenerative blower and the feed container **810** are positioned within the control room **120**. In some embodiments, the feeding subsystem **800** can also be used to dispense other substances (e.g., probiotics, sugars, etc.) that are in powder or particulate form. In some embodiments, the feeding subsystem **800** includes a main feeder to dispense food as described above, and a smaller version of the main feeder to dispense other substances. FIG. **8B** shows an exemplary feeding subsystem **850** including a container **860**, a weight sensor **870**, a motorized valve **880**, and a feeding tube **890** as described above.

(65) In some embodiments, a feeding strategy (e.g., frequency of feeding and feeding amount) is determined by the system control software in accordance with production protocols. In some embodiments, production algorithms determine an appropriate amount of food to dispense, taking into account the average size of organisms, quantity of organisms, growing stage, shedding stage, feed leftovers, etc. In some embodiments, the daily amount of food and frequency of feed (e.g., times per day) are determined by an algorithm operating on a remote server and are sent to a controller operating locally at the exemplary system to execute the feeding schedule. In some embodiments, the control software is programmed to instruct the water movement subsystem to allow the water in the tanks to be still for a set period after feed is dispensed to allow the shrimp to feed. In some embodiments, the period that elapses between the dispensing of the feed and the start of the circular water movement varies depending on factors such as the growing stage, amount of feed, etc. In some embodiments, following this period, the control software is programmed to instruct the water movement subsystem to activate the water jets thereof to cause water within the tanks to circulate, thereby allowing solid matter at the bottom of the tanks to be collected. In some embodiments, by collecting solid matter at a set interval following collection, the control software is able to become informed of the feeding leftovers after such a set interval, and to improve the feeding strategy on this basis.

(66) In some embodiments, the exemplary system includes a temperature control subsystem that is configured to maintain the water within the tanks at a suitable temperature. FIG. **9** illustrates an exemplary temperature control subsystem **900**. In some embodiments, a suitable temperature is a temperature in a range of between 28° C. and 33° C. In some embodiments, the temperature control subsystem **900** includes a heat exchanger **910**. In some embodiments, the heat exchanger **910** is positioned so that water caused to circulate around the tank by the water movement subsystem

passes through the heat exchanger and back to the tanks. In some embodiments, such as illustrated in FIG. 9, the temperature control subsystem **900** includes a pump **920** that is operative to pump water out of each tank, via tubing **930**, through the heat exchanger **910**, and back to each tank. In some embodiments, the pump **920** is a centrifugal pump. In some embodiments, the pump is a centrifugal pump **920** having a power of between 0.5 horsepower and 2 horsepower. In some embodiments, the pump **920** is a 1 horsepower centrifugal pump. In some embodiments, the heat exchanger **910** is controlled by the main controller of the exemplary system. In some embodiments, the main controller is operable to allow the suitable temperature to be configured.

(67) In some embodiments, the exemplary system includes a biofloc removal subsystem. In some embodiments, it is appropriate to ensure an adequate level of biofloc within the water tanks (e.g., to avoid the water tanks containing too much biofloc) because, while the organisms within the biofloc perform important functions as discussed above, the organisms also consume oxygen that is also needed by the shrimp or other organisms being raised within the water tanks.

(68) FIG. **10A** illustrates an exemplary biofloc removal subsystem **1000**. In some embodiments, the exemplary biofloc removal subsystem includes two biofloc removal devices **1010**, one for each water tank. In some embodiments, each biofloc removal device **1010** includes a decanter **1020** (e.g., an open-topped container configured to hold water) positioned within the corresponding water tank **310** such that the top of the decanter **1020** is above the water level of the corresponding water tank **310**. FIG. **10B** illustrates a perspective view of an exemplary decanter **1020**. In some embodiments, each decanter **1020** is chute-shaped and has a tapered bottom **1022**. In some embodiments, each biofloc removal device **1010** includes a water pump that is configured to remove water from the corresponding water tank **310** and pump the water into the top of the corresponding decanter **1020**. In some embodiments, the water pump is a submersible pump positioned within the corresponding water tank **310**. In some embodiments, as the decanter **1020** is positioned within the water tank **310**, any excess water positioned within the decanter **1020** overflows the decanter and returns to the water tank **310**. In some embodiments, water within the decanter **1020** is stationary and, as a result, biofloc within the decanter **1020** settles to the bottom **1022** of the decanter **1020**. In some embodiments, each biofloc removal device **1010** includes a drain pump **1030** that is operable to remove biofloc from the tapered bottom **1022** of the corresponding decanter **1020** via drain piping **1040**. In some embodiments, the drain pump **1030** is configured to turn on and off at periodic intervals. In some embodiments, the drain pump **1030** is configured to turn on and off based on the amount of biofloc within the decanter **1020** or within the water tank. In some embodiments, biofloc removed via the drain piping **1040** flows to a container **1050**. In some embodiments, each decanter **1020** includes a biofloc aeration arrangement including an aeration tube **1060** that is connected to the main aeration system discussed above, and is separated from the bottom of the decanter **1020** by a valve (e.g., an electric ball valve). In some embodiments, when the system controller determines that enough biofloc has been removed, the valve is opened to allow air from the aeration subsystem **400** or **450** to flow into the bottom of the decanter **1020** via the aeration tube **1060**. In some embodiments, when air flows in this manner, the solids at the bottom of the decanter **1020** (e.g., biofloc) mix with the water within the decanter **1020** and spill over the edge of the decanter **1020** and back into the water tank **310** in the same manner as described above for excess water. In some embodiments, removal of biofloc from the decanter **1020** in this manner prevents decomposition of biofloc within the decanter **1020**, which would be detrimental to the function of the system **100** (e.g., presence of decomposing matter would be detrimental to the health of organisms within the water tanks **310**).

(69) The exemplary embodiments described above produce shrimp of excellent quality in a sustainable manner and with minimal environmental impact through the use of biofloc together with optimizations in system processes such as feeding, aeration, heating, etc. The exemplary embodiments also enable production in a smaller amount of space than existing techniques. The exemplary embodiments also provide a modular system that can be replicated and scaled, and can

be deployed in different areas to produce shrimp locally to customers.

(70) Throughout the specification and claims, the following terms take the meanings explicitly associated herein, unless the context clearly dictates otherwise. The phrases “in one embodiment,” “in an embodiment,” and “in some embodiments” as used herein do not necessarily refer to the same embodiment(s), though they may. Furthermore, the phrases “in another embodiment” and “in some other embodiments” as used herein do not necessarily refer to a different embodiment, although they may. Thus, as described herein, various embodiments of the invention may be readily combined, without departing from the scope or spirit of the invention.

(71) As used herein, the term “based on” is not exclusive and allows for being based on additional factors not described, unless the context clearly dictates otherwise. In addition, throughout the specification, the meaning of “a,” “an,” and “the” include plural references. The meaning of “in” includes “in” and “on.”

(72) While a number of embodiments of the present invention have been described, it is understood that these embodiments are illustrative only, and not restrictive, and that many modifications may become apparent to those of ordinary skill in the art. For example, any dimensions discussed herein are provided as examples only, and are intended to be illustrative and not restrictive.

## Claims

1. A system, comprising: a housing, wherein the housing is an intermodal container; at least one water tank positioned within the housing, wherein an inside of each of the at least one water tank is sealed so as to prevent contact between a material forming the at least one water tank and water within the at least one water tank; a water movement subsystem operable to cause water to circulate within the at least one water tank, wherein the water movement system comprises: at least one baffle extending longitudinally along a portion of a center of the at least one water tank so as to divide the at least one water tank into a generally circular raceway, wherein the at least one baffle comprises a rough surface that is configured to allow crustaceans to attach to the rough surface; and at least one pump configured to cause water to circulate about the at least one baffle, wherein at least one of the at least one pump is positioned on the at least one baffle; an aeration subsystem configured to maintain an oxygenation level within the at least one water tank, wherein the oxygenation level is suitable for crustaceans to live within water in the at least one water tank; a temperature control subsystem operable to maintain a water temperature within the at least one water tank, wherein the water temperature is suitable for crustaceans to live within water in the at least one water tank; a water quality monitoring subsystem configured to monitor at least one water quality parameter of water within the at least one water tank, wherein the at least one water quality parameter includes a temperature, a dissolved oxygen concentration, a nitrogen concentration, a phosphate concentration, a pH, and a salinity; a feeding subsystem configured to dispense food into the at least one water tank, wherein the feeding subsystem comprises a feeding tube coupled to the aeration subsystem such that air provided by the aeration subsystem propels feed into the at least one water tank; a biofloc removal subsystem comprising a decanter configured to remove biofloc from the at least one water tank, wherein said decanter includes a biofloc aeration arrangement operable to cause air to flow into the bottom of the decanter when the biofloc aeration arrangement is activated, thereby to cause biofloc to mix within water within the decanter and to overflow into the at least one water tank, the biofloc aeration arrangement comprises: a valve that is operable to activate the aeration arrangement; and an aeration tube that is coupled to the valve and to the aeration subsystem; and a computer-operated control system configured to operate the water movement system, the aeration system, the temperature control subsystem, the water quality monitoring subsystem, and the feeding subsystem.

2. The system of claim 1, wherein the aeration subsystem comprises: at least one main air tube extending above each of the at least one water tank, and a plurality of branch tubes extending away

- from each of the at least one main air tube into water within the at least one water tank, wherein each of the plurality of branch tubes comprises aeration tubing.
3. The system of claim 2, wherein each of the plurality of branch tubes further includes polyvinyl carbonate tubing.
  4. The system of claim 3, wherein the polyvinyl carbonate tubing has a diameter of  $\frac{1}{2}$  inch.
  5. The system of claim 2, wherein the aeration tubing includes porous tubing having an average pore size that is in a range of from 0.001 inch to 0.004 inch.
  6. The system of claim 2, wherein the aeration subsystem comprises a main aeration subsystem and a secondary aeration subsystem.
  7. The system of claim 6, wherein the secondary aeration subsystem comprises at least one venturi eductor coupled to the water movement subsystem.
  8. The system of claim 1, wherein the feeding subsystem further includes: a scale; a feed container positioned on the scale; and a valve positioned at a bottom of the feed container and operable to allow feed to pass therethrough when the valve is in an open position, wherein the feeding tube is coupled to the aeration subsystem and to the valve such that the air provided by the aeration subsystem propels the feed passing through the valve into the at least one water tank.
  9. The system of claim 8, wherein the system includes one of the feeding subsystem for each of the at least one water tank.
  10. The system of claim 1, wherein the at least one water tank includes a first water tank positioned at a bottom of the housing and a second water tank positioned above the first water tank.
  11. The system of claim 10, further comprising a support structure configured to support weight of at least the second water tank.
  12. The system of claim 1, wherein each of the at least one water tank has a volume that is in a range of from 10 cubic meters to 50 cubic meters.
  13. The system of claim 1, further comprising a control room positioned at a first end of the housing.
  14. The system of claim 1, wherein the system is configured to support presence of biofloc within the at least one water tank.
  15. The system of claim 14, wherein the computer-operated control system is configured to support presence of biofloc within the at least one water tank by controlling a ratio of carbon to nitrogen within water within the at least one water tank.
  16. The system of claim 15, wherein the computer-operated control system is configured to support presence of biofloc within the at least one water tank by controlling the ratio of carbon to nitrogen within water within the at least one water tank to be in a range of from 12:1 to 20:1.
  17. The system of claim 1, wherein the biofloc removal subsystem comprises: the decanter positioned within the at least one water tank such that excess water within the decanter overflows into the at least one water tank, wherein the decanter is shaped such that biofloc within water in the decanter settles to a bottom of the decanter; a pump configured to pump water from the at least one water tank and into the decanter; and a drain pump operable to remove biofloc from the bottom of the decanter when the drain pump is activated.
  18. The system of claim 1, further comprising a drain subsystem operable to remove waste material from the at least one water tank.
  19. The system of claim 1, wherein each of the at least one water tank is sealed by a water-resistant paint.
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